

Optimization

3.2.3 Nelder–Mead simplex method

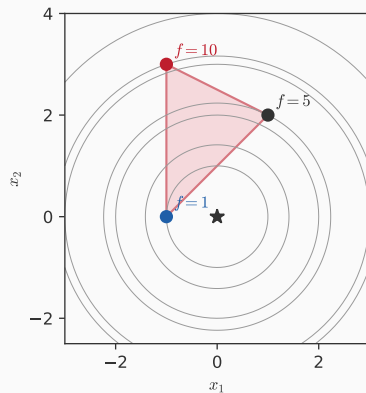
José Firmino Aguilar Madeira

From the grid to the simplex: can we use what we have already evaluated?

The grid (3.2.2) finds promising regions, but keeps spending evaluations where we already know it does not matter: as in random search, the points **do not use what has already happened**.

In 2D, with $f(x_1, x_2) = x_1^2 + x_2^2$, we evaluated three points:

$$f(-1, 0) = 1, \quad f(1, 2) = 5, \quad f(-1, 3) = 10.$$



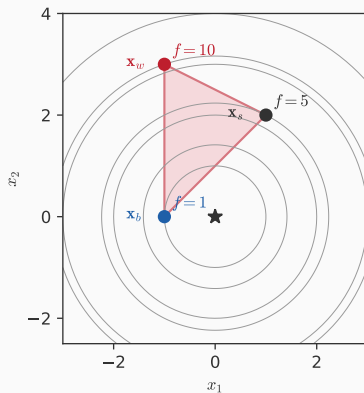
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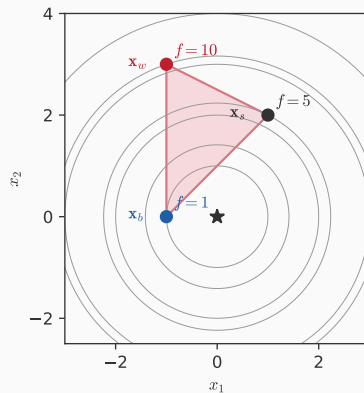
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If we only knew these three values, where would we try next?



Three points in 2D: reflection is born

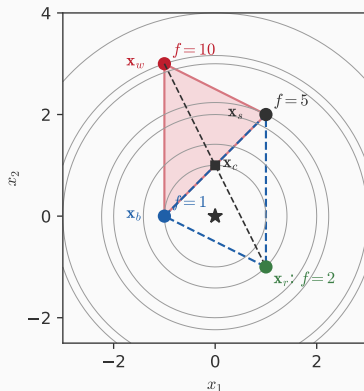
The natural answer: on the side opposite the worst point.

Centre of the other two: $\mathbf{x}_c = \frac{\mathbf{x}_b + \mathbf{x}_s}{2} = (0, 1)$.

Reflect the worst through $\mathbf{x}_c$:

$$\boxed{\mathbf{x}_r = \mathbf{x}_c + (\mathbf{x}_c - \mathbf{x}_w)} = (1, -1), \quad f(\mathbf{x}_r) = 2.$$

The worst ($f = 10$) is replaced by a point with $f = 2$:
the triangle **flips** away from the worst vertex.



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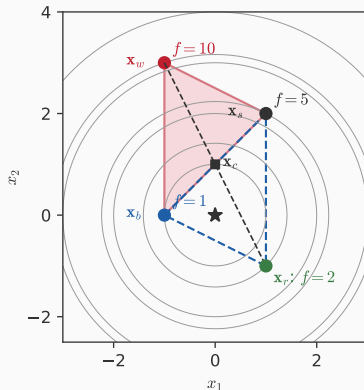
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The central idea of Nelder–Mead

With a few values of f , the **geometry** of the points already evaluated tells us where to try next. No gradient was computed.



The simplex in n dimensions

A **simplex** in $\mathbb{R}^n$ is the polytope with $n + 1$ vertices:

$n = 1$: segment; $n = 2$: triangle; $n = 3$: tetrahedron.

Nondegenerate = with nonzero (n -dimensional) volume (the vertices do not all lie in a hyperplane).

Nelder–Mead keeps $n + 1$ points and uses **only their values of f** . At each iteration it:

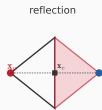
1. sorts the vertices: best $\mathbf{x}_b$, second worst $\mathbf{x}_s$, worst $\mathbf{x}_w$;
2. computes the centroid of the n vertices that are *not* the worst: $\mathbf{x}_c = \frac{1}{n} \sum_{i \neq w} \mathbf{x}_i$;
3. tries points on the line $\mathbf{x}_w \rightarrow \mathbf{x}_c$, on the other side.

Why it works

With $n + 1$ values of f the simplex “sees” the local slope without gradients: it moves away from the worst and heads in the direction in which f seems to decrease.

In 2D, $n = 2$: the centroid of the two best is the midpoint, as on the previous page.

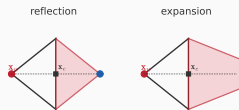
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All these points lie on the line defined by $\mathbf{x}_w$ and $\mathbf{x}_c$.

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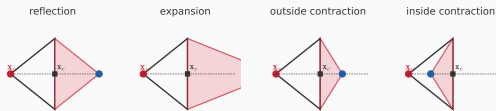
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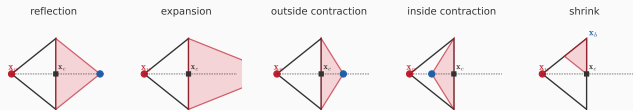
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- **Not good enough** ($f(x_r) \geq f(x_s)$)? Perhaps we went too far: outside **contraction**, between x_c and x_r :
 $x_{oc} = x_c + \rho(x_r - x_c)$; or inside, between x_c and x_w :
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- **Not even contraction works?** The simplex is too large or badly placed: **shrink** towards x_b , $x_i \leftarrow x_b + \sigma(x_i - x_b)$, $\sigma = \frac{1}{2}$.

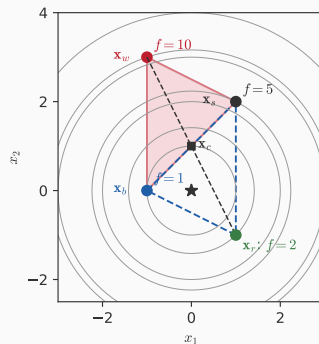
Operation	When
Expansion	$f(x_r) < f(x_b)$
Reflection accepted	$f(x_b) \leq f(x_r) < f(x_s)$
Outside contraction	$f(x_s) \leq f(x_r) < f(x_w)$
Inside contraction	$f(x_r) \geq f(x_w)$

$(\alpha, \gamma, \rho, \sigma) = (1, 2, \frac{1}{2}, \frac{1}{2})$: original coefficients (Nelder and Mead, 1965). Here α and γ are simplex coefficients, not the step α_k of 3.1.

One iteration by hand

$f(x_1, x_2) = x_1^2 + x_2^2$; the triangle from the start.

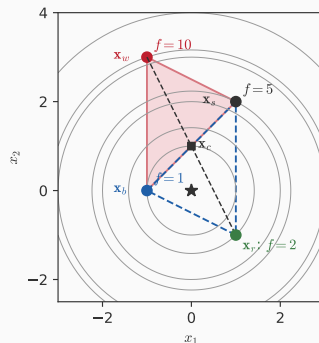
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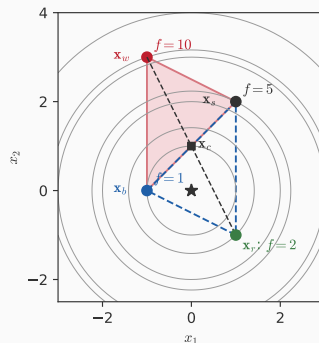
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 $\mathbf{x}_c = \frac{1}{2} [(-1, 0) + (1, 2)] = (0, 1)$.



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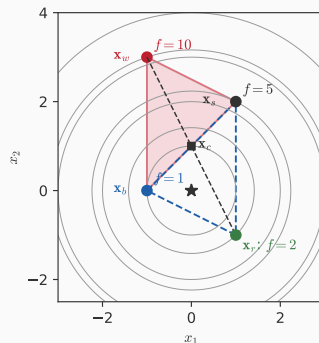
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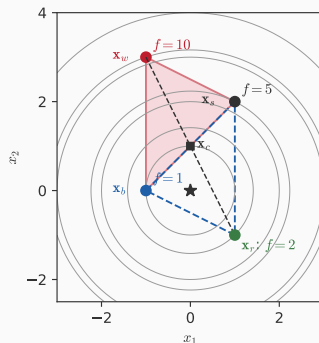
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 $f(\mathbf{x}_r) = 2$.
4. **Decide:** $f(\mathbf{x}_b) = 1 \leq f(\mathbf{x}_r) = 2 < f(\mathbf{x}_s) = 5 \Rightarrow$ reflection
accepted (it is not better than the best, so no expansion is
tried; it is better than the second worst, so no
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contraction).
5. **New simplex:** $\{(-1, 0), (1, -1), (1, 2)\}$, with $f = 1, 2, 5$.
It cost **one** evaluation of f . At the next iteration, the
worst is $(1, 2)$.



The new triangle already contains the minimum $(0, 0)$; the following iterations will shrink it around the minimum.

The algorithm

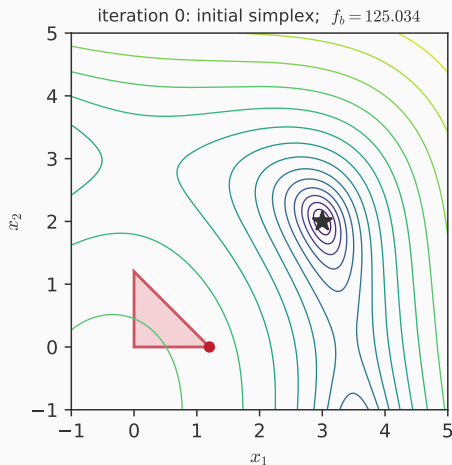
Nelder–Mead ($\alpha = 1$, $\gamma = 2$, $\rho = \frac{1}{2}$, $\sigma = \frac{1}{2}$)

```
1: choose  $n + 1$  vertices; evaluate  $f$ 
2: loop
3:   sort:  $f(\mathbf{x}_b) \leq \dots \leq f(\mathbf{x}_s) \leq f(\mathbf{x}_w)$ 
4:   stop if  $f(\mathbf{x}_w) - f(\mathbf{x}_b) \leq \varepsilon_f$  and  $\max_i \|\mathbf{x}_i - \mathbf{x}_b\| \leq \varepsilon_x$ 
5:    $\mathbf{x}_c \leftarrow \frac{1}{n} \sum_{i \neq w} \mathbf{x}_i$ ;  $\mathbf{x}_r \leftarrow \mathbf{x}_c + \alpha(\mathbf{x}_c - \mathbf{x}_w)$ 
6:   if  $f(\mathbf{x}_r) < f(\mathbf{x}_b)$  then ▷ expand
7:      $\mathbf{x}_e \leftarrow \mathbf{x}_c + \gamma(\mathbf{x}_r - \mathbf{x}_c)$ ;  $\mathbf{x}_w \leftarrow \mathbf{x}_e$  if  $f(\mathbf{x}_e) < f(\mathbf{x}_r)$ , else  $\mathbf{x}_r$ 
8:   else if  $f(\mathbf{x}_r) < f(\mathbf{x}_s)$  then  $\mathbf{x}_w \leftarrow \mathbf{x}_r$  ▷ accept
9:   else if  $f(\mathbf{x}_r) < f(\mathbf{x}_w)$  then ▷ outside contraction
10:     $\mathbf{x}_{oc} \leftarrow \mathbf{x}_c + \rho(\mathbf{x}_r - \mathbf{x}_c)$ 
11:    if  $f(\mathbf{x}_{oc}) \leq f(\mathbf{x}_r)$  then  $\mathbf{x}_w \leftarrow \mathbf{x}_{oc}$ 
12:    else shrink
13:    end if
14:  else ▷ inside contraction
15:     $\mathbf{x}_{ic} \leftarrow \mathbf{x}_c - \rho(\mathbf{x}_c - \mathbf{x}_w)$ 
16:    if  $f(\mathbf{x}_{ic}) < f(\mathbf{x}_w)$  then  $\mathbf{x}_w \leftarrow \mathbf{x}_{ic}$ 
17:    else shrink
18:    end if
19:  end if
20: shrink:  $\mathbf{x}_i \leftarrow \mathbf{x}_b + \sigma(\mathbf{x}_i - \mathbf{x}_b)$  for all  $i \neq b$ 
20: end loop
```

sort $\rightarrow$ centroid $\rightarrow$ reflect
excellent $\rightarrow$ expand; reasonable $\rightarrow$
accept;
bad $\rightarrow$ contract; still bad $\rightarrow$ shrink

- **Cost:** $n_f = 1\text{--}2$ per iteration; $n + 2$ if there is a shrink ($\mathbf{x}_r$, the contraction point and n new vertices).
- **Initial simplex:** $\mathbf{x}_0$ and $\mathbf{x}_0 + h_i \mathbf{e}_i$, $i = 1, \dots, n$, with $h_i = 0.05 \max(1, |\mathbf{x}_{0,i}|)$: relative increment, with an absolute minimum for zero components; in the examples: edges 1.2 (Himmelblau) and 0.5 (Rosenbrock).

Himmelblau from (0,0): the simplex in action

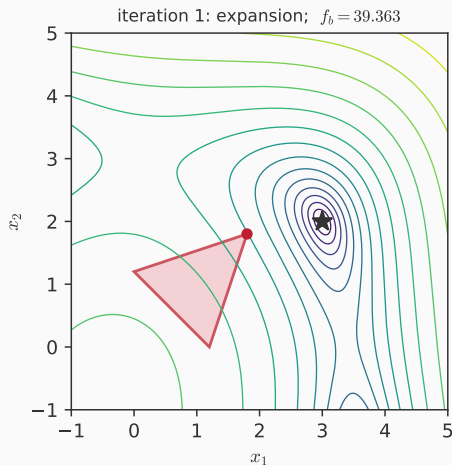


Initial simplex $(0,0)$, $(1.2,0)$, $(0,1.2)$.

k	operation	$\mathbf{x}_b$	$f(\mathbf{x}_b)$	$f(\mathbf{x}_w)$
0	initial	(1.200, 0.000)	125.0336	170.000

* expansion was tried ($f = 97.01$, worse); the reflected point was kept.

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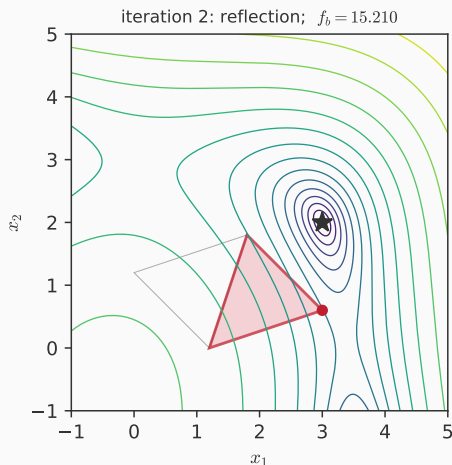


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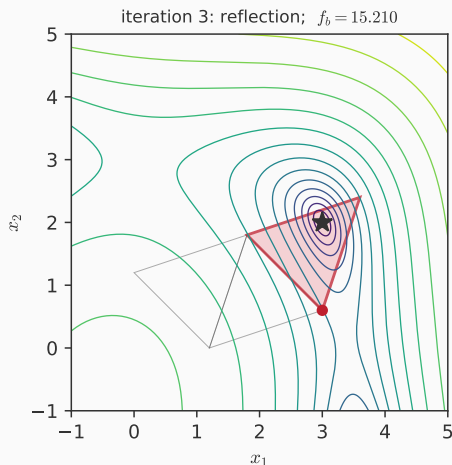


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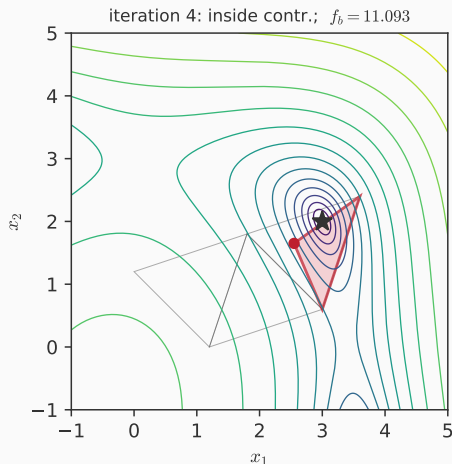


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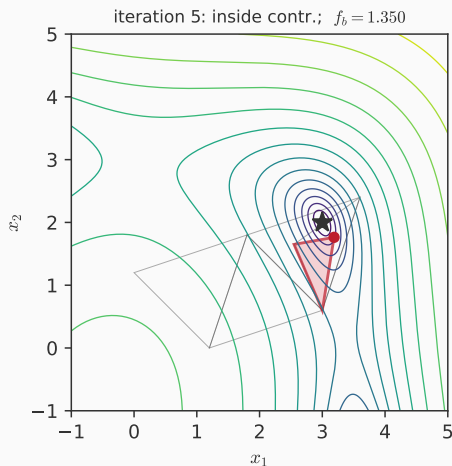


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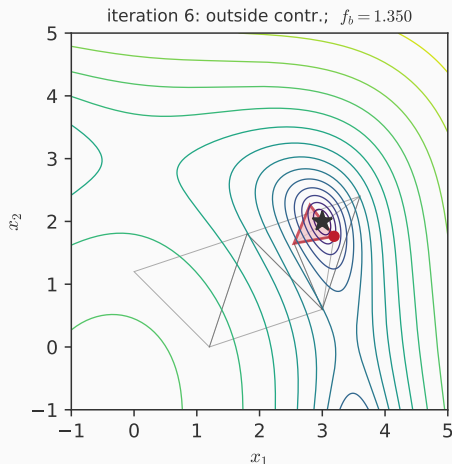


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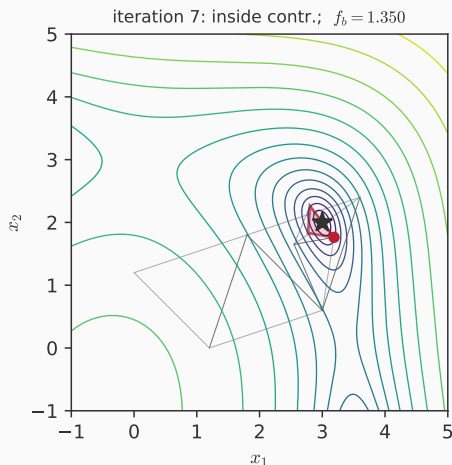


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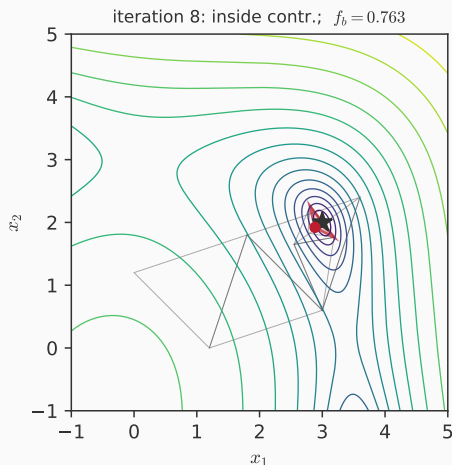


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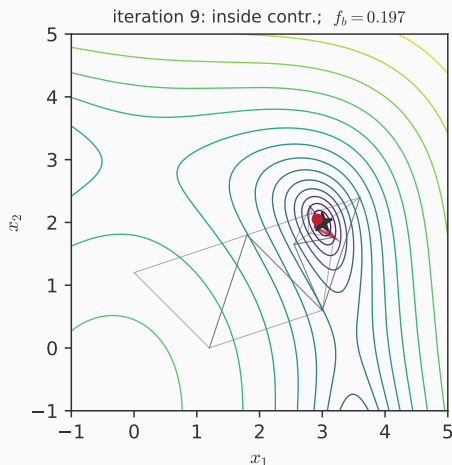


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* expansion was tried ($f = 97.01$, worse); the reflected point was kept.

Himmelblau from (0,0): the simplex in action

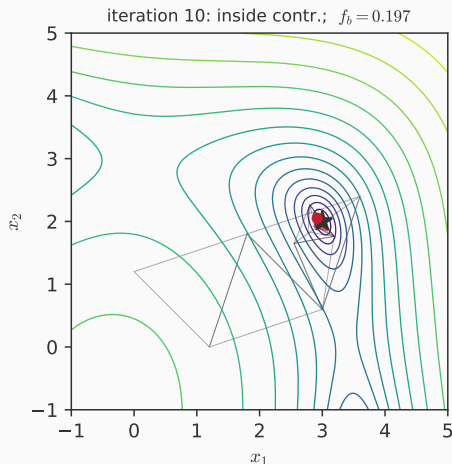


Initial simplex (0,0), (1.2,0), (0,1.2).

k	operation	$\mathbf{x}_b$	$f(\mathbf{x}_b)$	$f(\mathbf{x}_w)$
0	initial	(1.200, 0.000)	125.0336	170.000
1	expansion	(1.800, 1.800)	39.3632	126.954
2	reflection*	(3.000, 0.600)	15.2096	125.034
3	reflection	(3.000, 0.600)	15.2096	39.363
4	inside contr.	(2.550, 1.650)	11.0925	24.579
5	inside contr.	(3.188, 1.762)	1.3499	15.210
6	outside contr.	(3.188, 1.762)	1.3499	11.093
7	inside contr.	(3.188, 1.762)	1.3499	2.965
8	inside contr.	(2.884, 1.921)	0.7628	1.604
9	inside contr.	(2.919, 2.050)	0.1972	1.350

* expansion was tried ($f = 97.01$, worse); the reflected point was kept.

Himmelblau from (0,0): the simplex in action

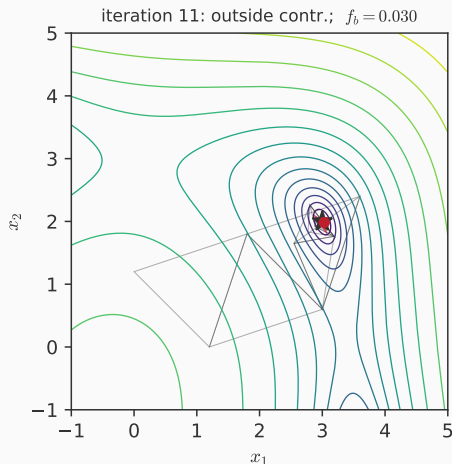


Initial simplex (0,0), (1.2,0), (0,1.2).

k	operation	$\mathbf{x}_b$	$f(\mathbf{x}_b)$	$f(\mathbf{x}_w)$
0	initial	(1.200, 0.000)	125.0336	170.000
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3	reflection	(3.000, 0.600)	15.2096	39.363
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8	inside contr.	(2.884, 1.921)	0.7628	1.604
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10	inside contr.	(2.919, 2.050)	0.1972	0.763

* expansion was tried ($f = 97.01$, worse); the reflected point was kept.

Himmelblau from (0,0): the simplex in action

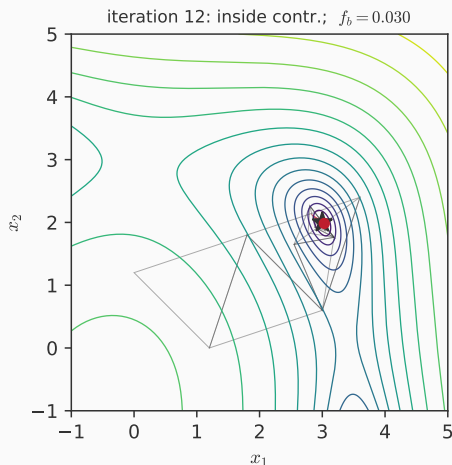


Initial simplex (0,0), (1.2,0), (0,1.2).

k	operation	$\mathbf{x}_b$	$f(\mathbf{x}_b)$	$f(\mathbf{x}_w)$
0	initial	(1.200, 0.000)	125.0336	170.000
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6	outside contr.	(3.188, 1.762)	1.3499	11.093
7	inside contr.	(3.188, 1.762)	1.3499	2.965
8	inside contr.	(2.884, 1.921)	0.7628	1.604
9	inside contr.	(2.919, 2.050)	0.1972	1.350
10	inside contr.	(2.919, 2.050)	0.1972	0.763
11	outside contr.	(3.031, 1.983)	0.0303	0.217

* expansion was tried ($f = 97.01$, worse); the reflected point was kept.

Himmelblau from (0,0): the simplex in action



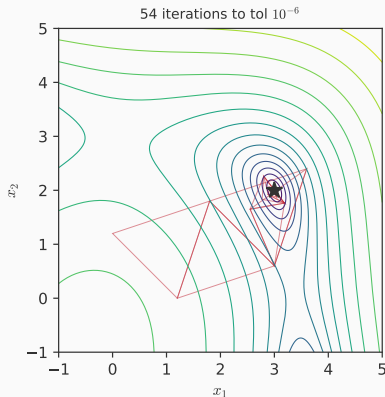
Initial simplex (0,0), (1.2,0), (0,1.2).

k	operation	$\mathbf{x}_b$	$f(\mathbf{x}_b)$	$f(\mathbf{x}_w)$
0	initial	(1.200, 0.000)	125.0336	170.000
1	expansion	(1.800, 1.800)	39.3632	126.954
2	reflection*	(3.000, 0.600)	15.2096	125.034
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6	outside contr.	(3.188, 1.762)	1.3499	11.093
7	inside contr.	(3.188, 1.762)	1.3499	2.965
8	inside contr.	(2.884, 1.921)	0.7628	1.604
9	inside contr.	(2.919, 2.050)	0.1972	1.350
10	inside contr.	(2.919, 2.050)	0.1972	0.763
11	outside contr.	(3.031, 1.983)	0.0303	0.217
12	inside contr.	(3.031, 1.983)	0.0303	0.197

* expansion was tried ($f = 97.01$, worse); the reflected point was kept.

The triangle **grows** when it finds descent (expansion), **flips** following the terrain (reflections) and **shrinks** near the minimum (contractions). $f(\mathbf{x}_w)$ decreases in jumps: it is the worst vertex that is replaced.

The complete path; on the computer



Converges to (3,2), $f \approx 0$: 54 iterations,
 $n_f = 106$ (tolerances 10^{-6}).

- Himmelblau has 4 minima: starting from (0,0) with the simplex to the right and upwards we reach (3,2); with the simplex (0,0), (-1.2,0), (0,-1.2) we reach (-2.81, 3.13). **Nelder-Mead is a local method**; the result depends on the initial simplex.

In MATLAB

`[x, fval] = fminsearch(f, x0)` is Nelder-Mead.
Useful options: `optimset('TolX',1e-6, 'TolFun',1e-6, 'Display','iter')`.

THE SAME IN PYTHON

```
r = scipy.optimize.minimize(f, x0,  
method='Nelder-Mead'); r.nfev counts the evaluations.
```

Implementations differ in the initial simplex and in the safeguards: same idea, different counts.

The price of Nelder–Mead

Advantages

- uses only values of f (order 0);
- simple; 1–2 evaluations per iteration ($n + 2$ with a shrink);
- adapts the position, size and shape of the simplex to the terrain;
- widely used in practice: `fminsearch`.

On Rosenbrock from $(-1.5, 2)$ it reaches $f < 10^{-4}$ with $n_f = 166$ (initial simplex with edges 0.5, the one used in the final comparison in 3.3.3); localized random search needed 975 [867, 1052] (median [Q1, Q3], 30 runs). On this problem, with these parameters.

Dimension: $n \lesssim 10$ is a common rule of thumb, not a boundary — it depends greatly on the problem.

Limitations

- local;
- depends on the initial simplex (on the same Rosenbrock, with 5% edges: $n_f = 139$);
- mostly used in low/moderate dimension: performance tends to degrade as n grows;
- it can stagnate: the simplex degenerates or shrinks away from the minimum;
- Nelder–Mead has no general convergence guarantee: there are smooth convex functions on which it stops at a nonstationary point (McKinnon, 1998);
- noise can disturb the ordering of the vertices.

Derivative-free $\neq$ robust to noise.

KEY TAKEAWAYS — 3.2.3

1. Nelder–Mead uses a simplex of $n + 1$ points and only values of f .
2. At each iteration it identifies the worst vertex and searches on the opposite side: reflection.
3. The quality of the reflection decides: expand, accept, contract or shrink.
4. The simplex adapts position, size and shape to the terrain, but the method is local. Nelder–Mead has no general convergence guarantee: there are smooth convex functions on which it stops at a nonstationary point (McKinnon, 1998).
5. Using only values of f does not mean being robust to noise.

Next (3.2.4): the simplex changes shape according to the values of f . What if, instead, we use a fixed pattern of points around the current point — the vertices of a box — and only move or shrink it? — Box's method.

References and resources

References Full list: *Course bibliography*, at the end of the course PDF.

- Nelder, J. A., Mead, R., *A simplex method for function minimization*, The Computer Journal 7(4):308–313, 1965.
- McKinnon, K. I. M., *Convergence of the Nelder–Mead simplex method to a nonstationary point*, SIAM Journal on Optimization 9(1):148–158, 1998.
- Conn, A. R., Scheinberg, K., Vicente, L. N., *Introduction to Derivative-Free Optimization*, SIAM (MPS-SIAM Series on Optimization 8), 2009 — Ch. 8.

Resources

- MATLAB: `fminsearch`, `optimset('TolX',..., 'TolFun',..., 'Display', 'iter')`.
- Python: `scipy.optimize.minimize(f, x0, method='Nelder-Mead')(r.nfev)`.